

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (IT) Examination

April 2013

MA206 Probability, Statistics & Numerical Analysis

Date: 26.04.2013, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION I

- Q 1 (a) A sample of $n = 8$ scores has a mean of $AM = 12$. A new score of $x = 3$ is added to the sample. Answer the following questions 02

- i. The $\sum x$ for new sample is _____.
- ii. The mean of new sample is _____.

- (b) What is the probability that the 2 cards drawn at random from an ordinary deck will be all spades? 01

- (c) Let S be a sample space and a probability function $P(\cdot)$ defined on subsets of S . Let E and F be any two events on S , State whether following are true or false. 03

- i. $P(E^c) = 1 - P(E)$
- ii. $P(E \cup F) = P(E) + P(F) + P(E \cap F)$
- iii. $P(F \cap E^c) = P(F) - P(E \cap F)$

- (d) Let the n pairs $(x_i, y_i), i = 1, 2, \dots, n$ denote the ranks associated with some characteristics under study, then rank correlation coefficient is given by _____ 01

- Q 2 (a) Number of software's validation procedure done by 150 employees in a software development company is given in frequency table below. 04

Number of Software validation procedure	0-15	15-30	30-45	45-60	60-75	75-90	90-105
No of employees:	8	26	30	45	20	17	4

- i. Construct histogram of the frequency distribution
- ii. Construct cumulative frequency distribution

- (b) Write three important properties of Arithmetic Mean 03

- (c) A computer manager needs to know how efficiency of her new computer program depends on the size of incoming data. Efficiency will be measured by the number of 07

processed requests per hour. Applying the program to data sets of different sizes, she gets the following results,

Data size (gigabytes), x	6	7	7	8	10	10	15
Processed requests, y	40	55	50	41	17	26	16

- Explain type of relationship between variables using scatter diagram.
- Compute the Karl Pearson correlation coefficient between variables.

OR

- Q 2 (a)** A Senior manager of a software development company has collected the following data on experience of the employee and time required rectifying errors in software. 07

Employee	1	2	3	4	5	6	7	8
Experience (year)	16	12	18	4	3	10	5	12
Time(minute)	30	45	20	58	68	49	52	42

Obtain least square line and estimate time (in minute) required rectifying errors in software by Employee who has 7 years of experience.

- (b)** Consider the following data. 07

14	19	24	19	16	20	24	20	16	16
21	22	24	18	17	23	26	22	16	12
23	25	25	19	18	16	15	24	25	19
21	16	19	21	23	20	22	22	24	20

- Construct a frequency distribution using classes: 12-14, 15-17, 18-20, 21-23, and 24-26
- Compute the arithmetic mean and standard deviation of the frequency distribution obtained in question above

- Q 3 (a)** Define terms with illustration: Random Experiment, Sample space, Mutually exclusive Events 06

- (b)** The time (in hours) required to repair a CPU is a random variable with probability density function given by $f(x) = \lambda e^{-\lambda x}$, $x > 0, \lambda > 0$. What is the probability that a repair time is below 2 hours? If $\lambda = \frac{1}{2}$ 04

- (c)** If random variable X has Normal distribution with parameters $\mu = 3$ and $\sigma^2 = 9$ Find $P(2 < X < 5)$. The following table gives $P(Z \leq z)$ 04

Z	0	3	6	7
0.3	0.6179	0.6293	0.6406	0.6443
0.6	0.7257	0.7357	0.7454	0.7486
1.0	0.8413	0.8485	0.8554	0.8577

OR

- Q 3 (a) A communication system consists of 6 components, each of which will independently function with probability $p = 1/4$. 04

Use Binomial probability function and find the probability that

- i. At least two components functioning
 - ii. Exactly three components functioning
- (b) Among employees of a certain firm, 70% know C/C++, 60% know Fortran, and 50% know both languages. What proportion of programmers, 06

i. Does not know Fortran?	ii. Does not know Fortran and C/C++?
iii. Knows C/C++ but not Fortran?	iv. Knows Fortran but not C/C++?

- (c) Out of six computer chips, two are defective. If two chips are randomly chosen for testing (without replacement), compute the probability that both of them are defective. List all the outcomes in the sample space. 04

SECTION II

- Q 4 (a) The function $f(x)$ and n^{th} degree polynomial $P_n(x)$ have the same values at tabulated values of x . The values of x are equidistant. Then $\Delta^{n+1}P_n(x) = 0$ (True/False) 01

- (b) If the true value is $\frac{10}{3}$ and approximate value is 3.333, answer following questions. 01

- i. The relative error is _____.
- ii. The number of significant digits in approximate number is _____.

- (c) Write the normal equations to obtain a curve fit of the type $y = ae^{bx}$ to data 02

- (d) The composite Newton-Cotes formula based on quadratic interpolating polynomial is known as _____ (Trapezoidal rule, Simpson's 1/3 rule, Simpson 3/8 rule) 01

- (e) If $x = \alpha$, $a < \alpha < b$ be a root of algebraic equation $f(x) = 0$ which is continuous in the interval $[a, b]$. If $f(a)$ and $f(b)$ are of opposite sign, answer the following questions 02

- i. The approximate value of root α using Bisection Method is given by formula $x_0 =$ _____.
- ii. If x_0 is the approximate root of α obtained by Bisection Method then the next approximate value of root lies in the interval $[a, x_0]$ if _____.

- Q 5 (a) Define relative error. Given that a number 6.78 is approximate value of x , and relative error in x is not given to be more than 5 %, what should be the range of actual value of x . 04

- (b) Write the Lagrangian interpolating polynomial that passes through each point 04

x	-1.1	2.1	3.5
y	1.3	2.5	-1.6

- (c) Fit an appropriate straight line y on x to the data given below 06

x	1	2	3	4	5	6
y	5.04	8.12	10.64	13.18	16.20	20.04

OR

- Q 5 (a) Define Finite Differences. Construct the finite differences of the function $f(x) = x^3$ 04

- (b) The table below gives the value of $\tan x$ for $0.10 \leq x \leq 0.30$ 04

x	0.10	0.15	0.20	0.25	0.30
y = tan x	0.1003	0.1511	0.2027	0.2553	0.3093

Use Newton's Difference Interpolation formula and find $\tan 0.12$

- (c) Fit an appropriate exponential curve $y = ae^{bx}$ to the data given below 06

x	77	100	185	239	285
y	2.4	3.4	7.0	11.1	19.6

- Q 6 (a) Find approximate root of $f(x) = 0$ where $f(x) = e^x - 4x^2$ by Newton Raphson method, beginning with $x_0 = 1$. Perform four iterations only. 04

- (b) Evaluate $\int_1^3 \ln(x+2) dx$ numerically using Simpson's $\frac{1}{3}$ rule with six sub intervals 06

(Note that $\ln(x) = \log_e(x)$). Find an error in integration by direct evaluation.

- (c) Given the differential equation $\frac{dy}{dx} = -y$, $y(0) = 1$, Solve the equation using Euler's method and determine $y(0.04)$. Take $h = 0.01$ 04

OR

- Q 6 (a) Evaluate numerically $\int_{-1}^1 \frac{1}{1+x^2} dx$ using Trapezoidal formula with $n = 7$. Find an error in integration by direct evaluation. 07

- (b) Given $\frac{dy}{dx} = -xy^2$, $y(2) = 1$ find numerically $y(2.2)$, correct to four decimal places using fourth order Runge - Kutta method, with $h = 0.1$, 07
